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Subject : OOPS lab

Labtask 7

Question 1:

Code:

Q1.cpp > ...

```cpp
// labtask7
// Q1

#include<iostream>
#include<cstring>
using namespace std;

class Student
{
    string name;
    int roll_no;
    int marks;

    public:
    // Default constructor
    Student()
    {
        name = "Unknown name";
        roll_no =0;
        marks = 0;

    }

    // parameterized constructor
    Student(string n, int r, int m)
    {
        name = n;
        roll_no = r;
        marks = m;

    }

```

```
    void display()
    {
        cout<<"name : "<<name<<endl;
        cout<<"Roll Number : "<<roll_no<<endl;
        cout<<"Marks : "<<marks<<endl;
    }

};

int main()
{
    cout<<"----Default constructor called ----"<<endl;
    Student s1;
    s1.display();
    cout<<endl;

    cout<<"----Parameterized constructor called ----"<<endl;
    Student s2("Ali", 1234, 80);
    s2.display();
    cout<<endl;

}
```

Output:

```
----Default constructor called ----
name : Unknown name
Roll Number : 0
Marks : 0

----Parameterized constructor called ----
name : Ali
Roll Number : 1234
Marks : 80
```

Question 2:

Code:

```cpp
// labtask7
// Q1

#include <iostream>
#include <cstring>
using namespace std;

class Result
{
    string name;
    int roll_no;
    int marks[5];

public:
    // Default constructor
    Result()
    {
        name = "Unknown name";
        roll_no = 0;
        for (int i = 0; i < 5; i++)
        {
            marks[i] = 0;
        }
    }

    // parameterized constructor
    Result(string n, int r, int m[]) : name(n) , roll_no(r)
    {

        for (int i = 0; i < 5; i++)
        {
            marks[i] = m[i];
        }
    }
```

```
    float average()
    {
        float sum = 0.0;
        for (int i = 0; i < 5; i++)
        {
            sum += marks[i];
        }

        float avg = sum / (float)5;
        return avg;
    }

    void display();
};

// out-of-line function definition
void Result::display()
{
    cout << "name : " << name << endl;
    cout << "Roll Number : " << roll_no << endl;
    cout << "Marks : ";

    for (int i = 0; i < 5; i++)
    {
        cout << marks[i] << " ";
    }
    cout << endl;

    float avg = average();
```

```
    }
    cout << endl;

    float avg = average();

    if (avg >= 50)
    {
        cout << " Status : Pass" << endl;
    }
    else
    {
        cout << "Status : Fail" << endl;
    }
}

int main()
{
    //----Default constructor called ----
    Result s1;
    cout << "-----Displaying result---" << endl;
    s1.display();
    cout << endl;

    //----Parameterized constructor called ----
    int arr[5] = {60, 70, 80, 90, 50};
    Result s2("Ali", 1234, arr);
    cout << "-----Displaying result---" << endl;
    s2.display();
    cout << endl;

    return 0;
}
```

Output:

```
-----Displaying result---
name : Unknown name
Roll Number : 0
Marks : 0 0 0 0 0
Status : Fail

-----Displaying result---
name : Ali
Roll Number : 1234
Marks : 60 70 80 90 50
 Status : Pass
```

Question 3:

Header File:

Q3_folder > class_header.h > ...

```cpp
#include <iostream>
#include <cstring>

using namespace std;

class Employee
{
    string name;
    double basic_salary;
    int ID;

    public :

    Employee();
    Employee(string n, double bs, int id);
    double bonus();
    void display();

};

```

CPP File:

Q3_folder > class_header.cpp > ...

```cpp
#include <iostream>
#include <cstring>
#include "class_header.h"
using namespace std;

// default constructor
Employee::Employee() : name("Unknown"), basic_salary(0), ID(0) {}

// parameterized constructor
Employee::Employee(string n, double bs, int id) : name(n), basic_salary(bs), ID(id) {}

// member function
double Employee::bonus()
{
    if (basic_salary > 0)
    {
        int bonus = basic_salary * 10 / 100;
        basic_salary += bonus;
        return bonus;
    }

    // in case of default constructor
    return 0.0;

}
// member function
void Employee::display()
{
    cout << "Employee name : " << name << endl;
    cout << "Employee Basic salary: " << basic_salary << endl;
    cout << "Employee ID : " << ID << endl;
    cout << "Bonus amount : " << bonus()<<endl;
    cout << "Employee Basic salary after 10% bonus : " << basic_salary << endl;
}
```

Main cpp file :

Q3_folder > main.cpp > main()

```cpp
// labtask7
// Q1

#include <iostream>
#include <cstring>
#include"class_header.h"
using namespace std;

int main()
{
    Employee e1;
    Employee e2("Ali" , 1234, 30);

    cout<<"----Details of employee 1----"<<endl;
    e1.display();
    cout<<"----Details of employee 2----"<<endl;
    e2.display();

}
```

Output:

```
syeda-areej-e-fatima@velanora:~/Sem_2/OOPs_lab/lab_
----Details of employee 1----
Employee name : Unknown
Employee Basic salary: 0
Employee ID : 0
Bonus amount : 0
Employee Basic salary after 10% bonus : 0
----Details of employee 1----
Employee name : Ali
Employee Basic salary: 1234
Employee ID : 30
Bonus amount : 123
Employee Basic salary after 10% bonus : 1357
syeda-areej-e-fatima@velanora:~/Sem_2/OOPs_lab/lab_
```